SRUTHI B

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ABOUT ME

Final-year Electronics and Communication Engineering student with a keen interest in Digital Electronics, VLSI, and hardware design. Experienced in developing FPGA-based digital systems using Verilog HDL and working with electronic circuit simulation tools. Seeking an internship opportunity to apply my technical knowledge and gain practical exposure to electronics and hardware development.

EDUCATION

B.E - Final year Electronics and Communication Engineering at Knowledge Institute of Technology **2023-2027 | 8.86 Cgpa**
12th - Completed at St. Mary's girls higher secondary school, Salem. **2022-2023 | 86.5**
10th - Completed at St. Mary's girls higher secondary school, Salem. **2020-2021**

SKILLS

HDL : Verilog, SystemVerilog basics
Core areas: Digital Electronics, Low power VLSI, FPGA based digital system design, ADC/DAC, DFT basics
FPGA boards: Intel DE10-Lite, Xilinx Spartan-7, Zynq Board
Tools : Xilinx Vivado, Intel Quartus Prime, LTspice, ModelSim, EDA playground, KiCad
Programming : Java, C

PROJECTS

1. Vending Machine Controller on Intel DE10-Lite

- Designed an FSM-based vending machine controller using Verilog HDL on the Intel DE10-Lite FPGA.
- Implemented ₹10/₹20/₹30/₹40 product selection, coin insertion, balance tracking, purchase, and refund operations using switches and push buttons.
- Verified the design through simulation in Intel Quartus Prime and validated the controller on the DE10-Lite using LEDs and a 7-segment display.

2. ALU on DE10-Lite

- Designed an ALU using Verilog HDL and implemented it on the Intel DE10-Lite FPGA.
- Implemented addition, subtraction, multiplication, and division, and verified the design using Intel Quartus Prime.

3. Energy Recovery Circuit using Adiabatic Logic

- Designed and simulated an Improved Diode-Free Adiabatic Logic (IDFAL) circuit using LTspice to reduce dynamic power consumption through energy recovery.
- Compared its energy recovery and power consumption with a conventional CMOS circuit.

4. IoT Vaccine Monitoring System

- Developed an IoT-based vaccine monitoring system to monitor temperature and maintain safe storage conditions.
- Automated Peltier cooling when the temperature exceeded the threshold and enabled real-time temperature alerts.

INTERNSHIP

Embedded System Trainee | Sans innovation - 2 weeks

- Learned the fundamentals of embedded systems and basic electronic circuits.
- Gained knowledge of rectifiers, AC-DC power conversion, and embedded devices such as smart wearables.

Internet of Things Intern | Government ITI - 2 weeks

- Gained hands-on experience with Arduino and IoT sensors.
- Learned sensor interfacing, data monitoring, and basic IoT applications through practical sessions.

ACHIEVEMENTS & PRESENTATION

- Research paper presented on "Energy Recovery Circuit with IDFAL" at the 5th International Conference (ICOST'26), Annapoorana Engineering College.
- First Prize - Technical quiz event at Dhirajlal Gandhi College of Technology.
- Paper Presentation – Electric Harvesting Tiles, Kongu Engineering College.
- Paper Presentation – Vibewave Intercom, Paavai Engineering College.

CERTIFICATIONS

- Completed Low power VLSI and FPGA based Digital system design from GUVI HCL.
- Completed Advanced digital system design using Verilog and Verification with System Verilog from GUVI HCL.